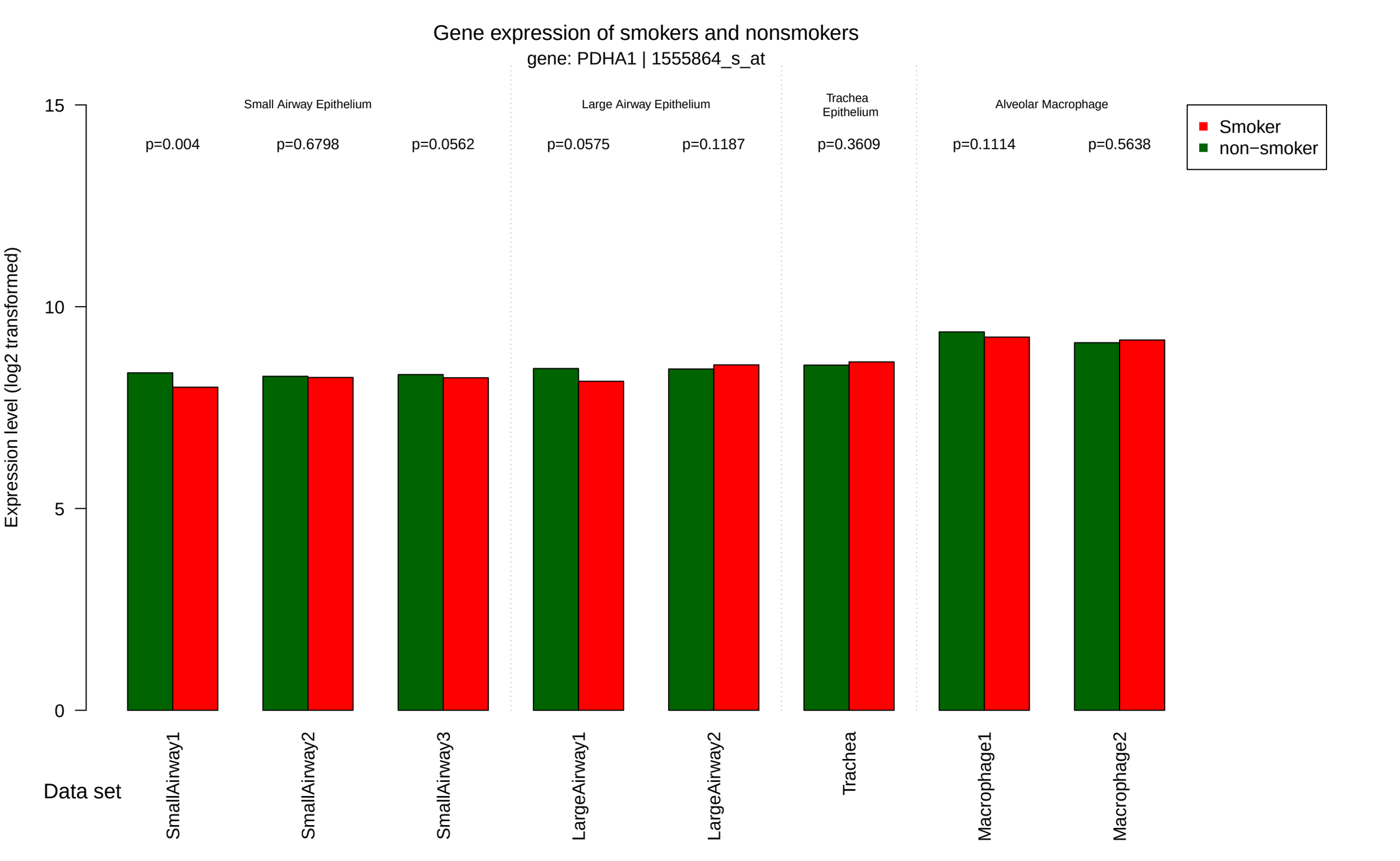
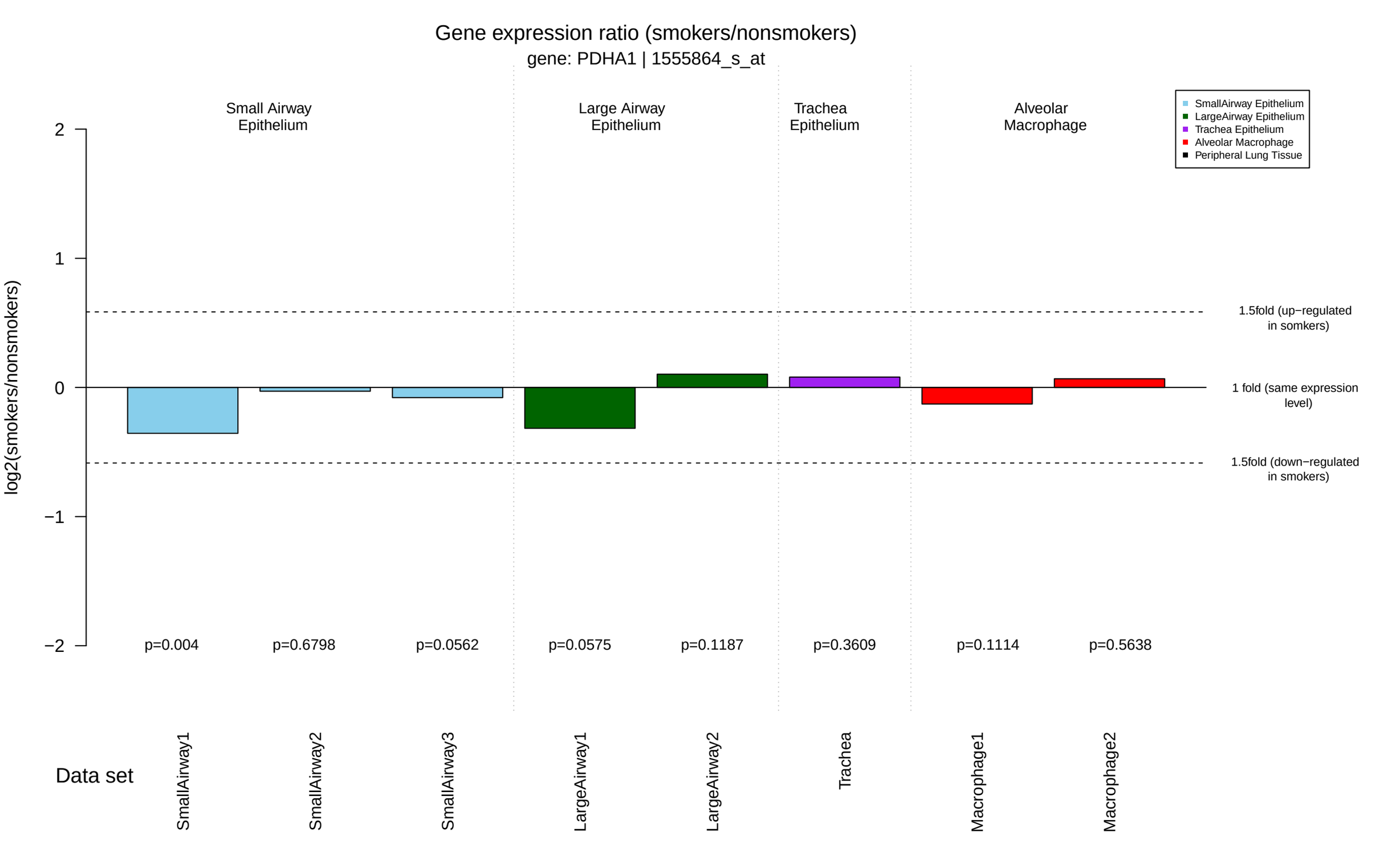
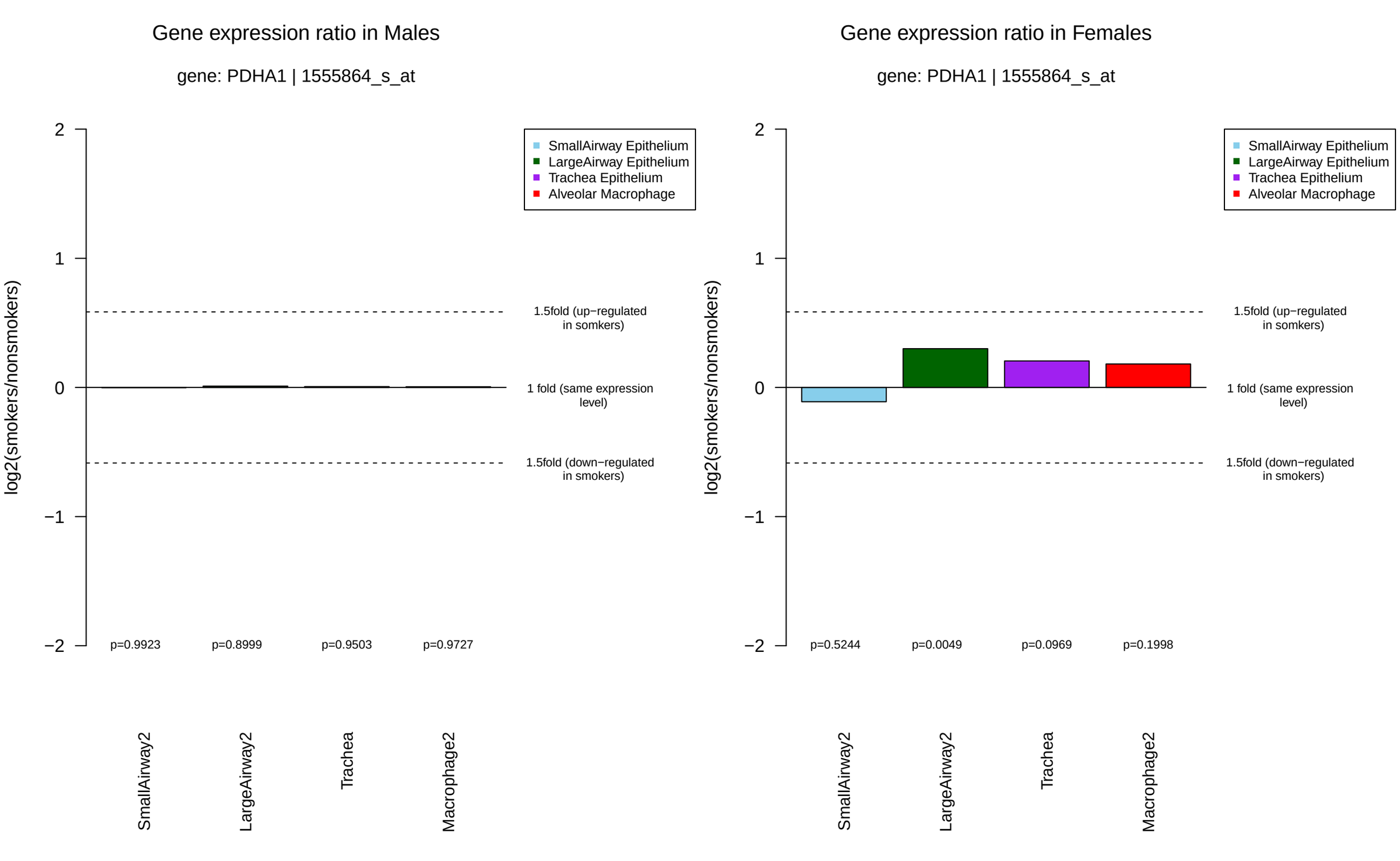
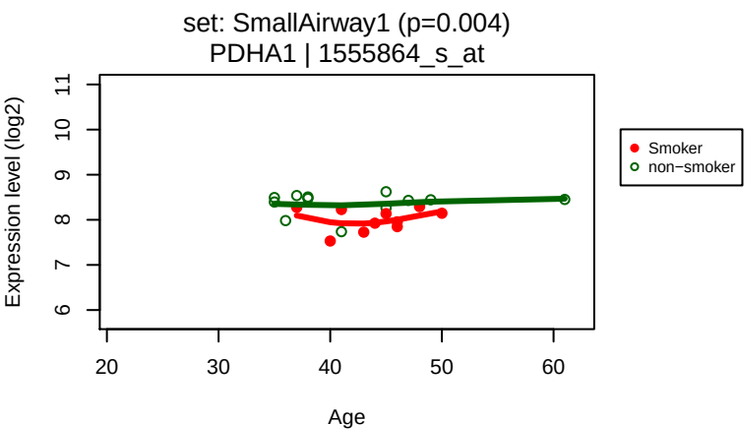
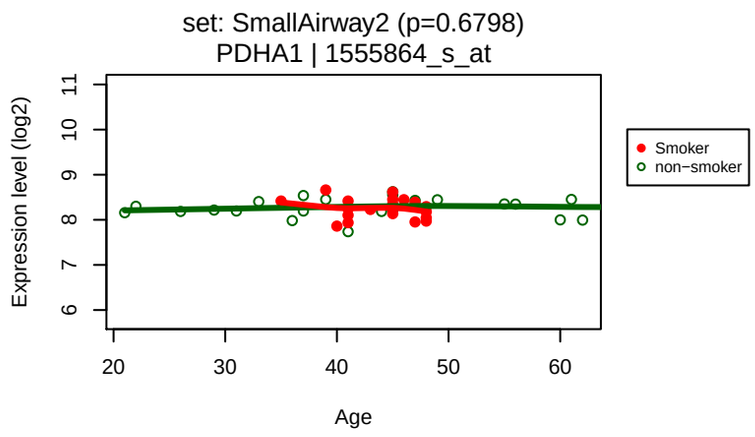
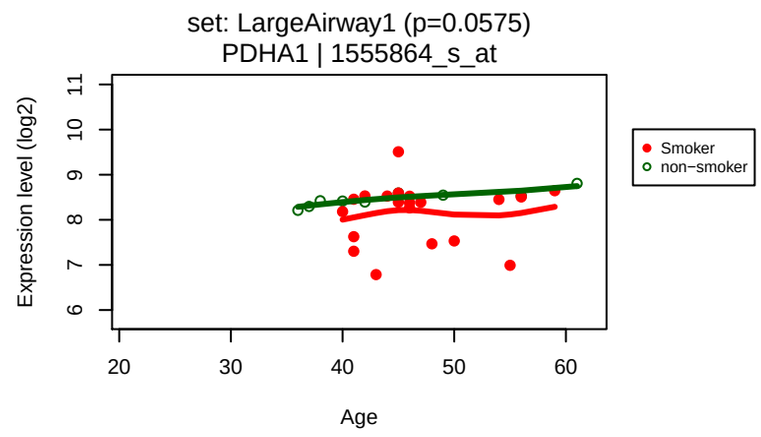

Gene expression of smokers and nonsmokers
gene: PDHA1 | 1555864_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.004
p=0.6798
p=0.0562
p=0.0575
p=0.1187
p=0.3609
p=0.1114
p=0.5638
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

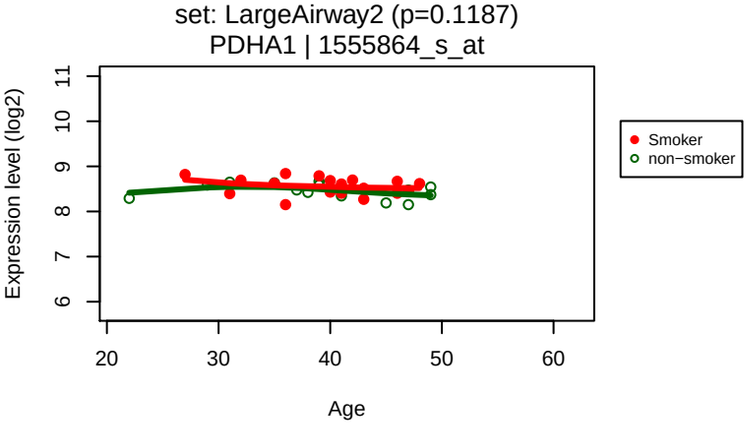

Gene expression ratio (smokers/nonsmokers)
gene: PDHA1 | 1555864_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.004
p=0.6798
p=0.0562
p=0.0575
p=0.1187
p=0.3609
p=0.1114
p=0.5638
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

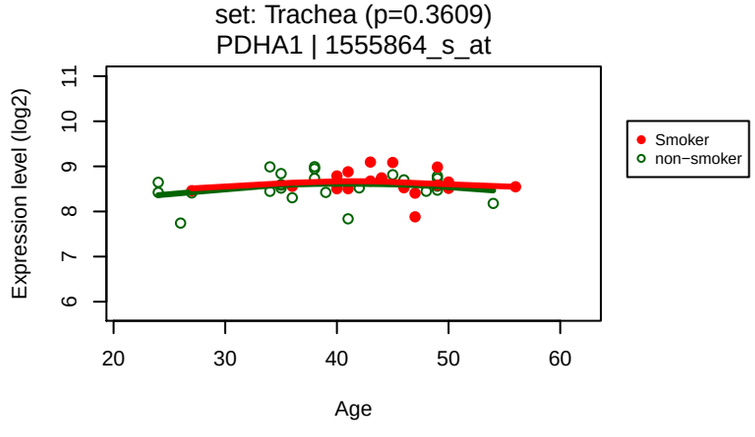

Gene expression ratio in Males
gene: PDHA1 | 1555864_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9923
p=0.8999
p=0.9503
p=0.9727
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDHA1 | 1555864_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5244
p=0.0049
p=0.0969
p=0.1998
SmallAirway2
LargeAirway2
Trachea
Macrophage2

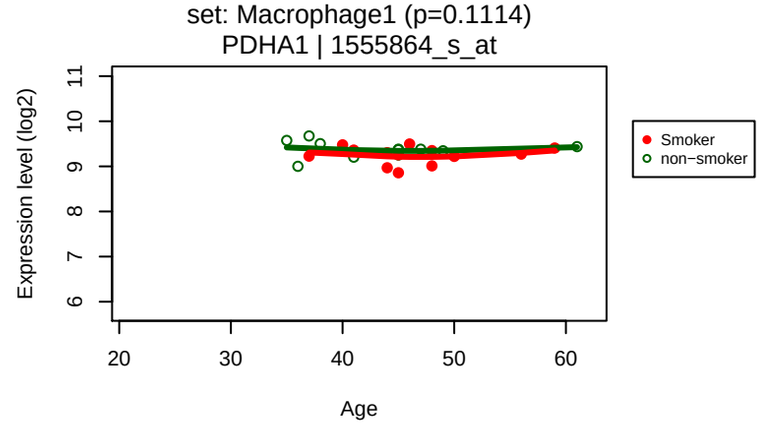

set: SmallAirway1 (p=0.004)
PDHA1 | 1555864_s_at
Expression level (log2)
Age
Smoker
non−smoker

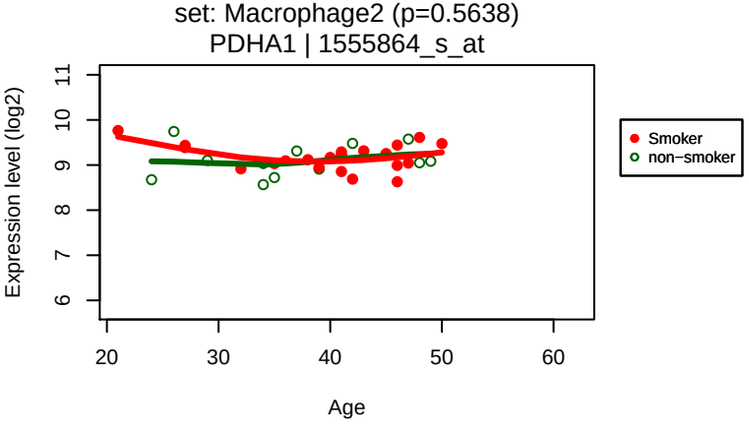

set: SmallAirway2 (p=0.6798)
PDHA1 | 1555864_s_at
Expression level (log2)
Age
Smoker
non−smoker

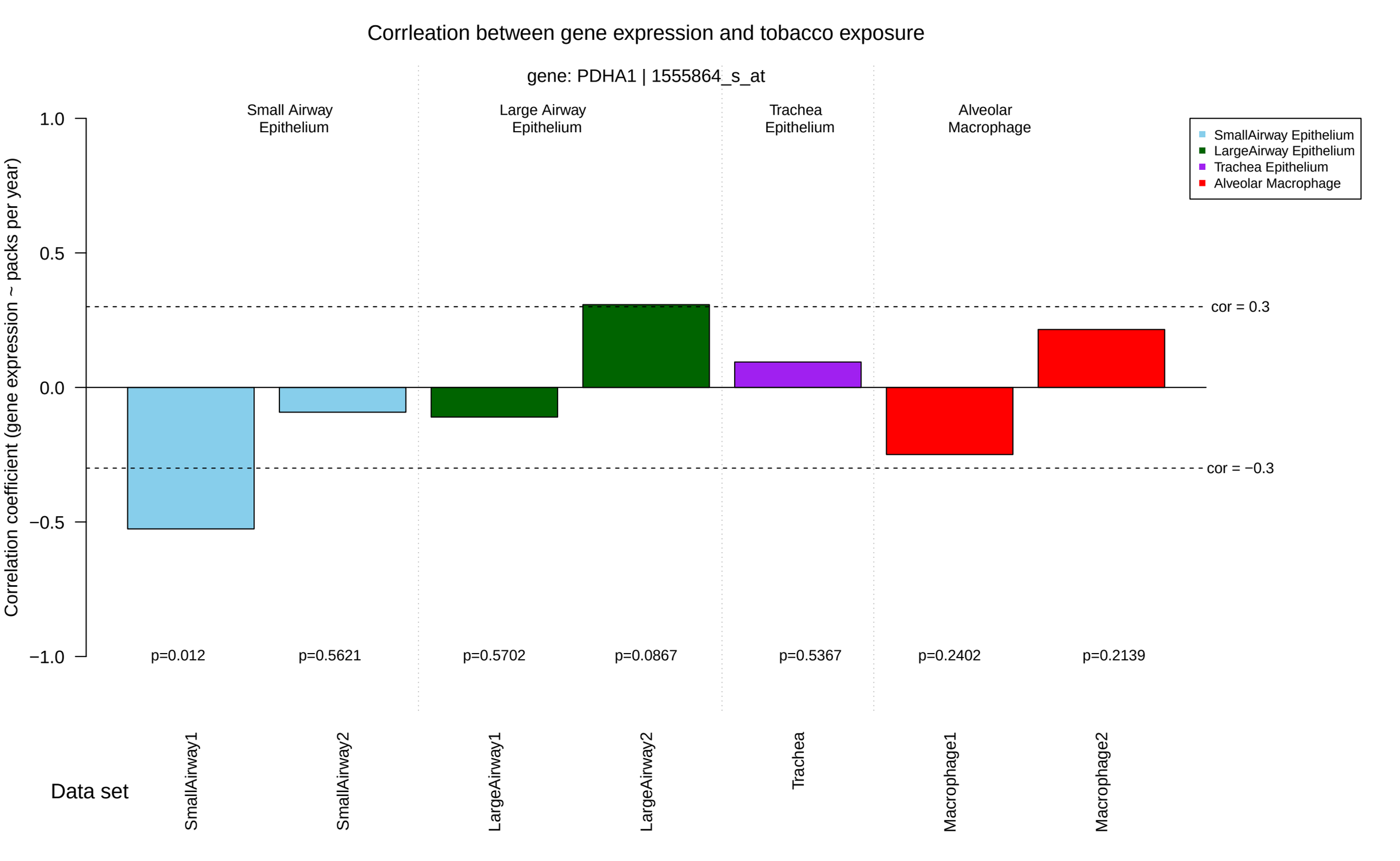

set: LargeAirway1 (p=0.0575)
PDHA1 | 1555864_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1187)
PDHA1 | 1555864_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3609)
PDHA1 | 1555864_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1114)
PDHA1 | 1555864_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5638)
PDHA1 | 1555864_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDHA1 | 1555864_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.012
p=0.5621
p=0.5702
p=0.0867
p=0.5367
p=0.2402
p=0.2139
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2